

Agenda

- Recap
- Intro to Multivariable Calculus
- Partial derivatives
- Intro to gradients
 - Representation of gradients
 - Geometric interpretation of gradients
- Computing gradient of any function
- Computing partial derivatives code
- Intuition of Gradient descent
- Generalization of gradient descent

Detailed Recap.

1] Derivatives: $f(x) \rightarrow$ continuous & Differentiable.

$$\frac{d}{dx} f(x) = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

2] Maxima Minima: $f(x) \rightarrow$ continuous & Differentiable.

To find candidate points for maxima & minima:

Find all x such that $f'(x) = 0$.

If: $f''(x) > 0$: Minima

$f''(x) < 0$: Maxima.

3] Rules:

(a) Linearity $\rightarrow$ If $h(x) = a \cdot f(x) + b \cdot g(x)$ then $h'(x) = a f'(x) + b g'(x)$

(b) Product $\rightarrow$ If $h(x) = f(x) \cdot g(x)$ then $h'(x) = f'(x)g(x) + g'(x)f(x)$

(c) Quotient $\rightarrow$ If $h(x) = \frac{f(x)}{g(x)}$ then $h'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{(g(x))^2}$

(d) Chain $\rightarrow$ If $h(x) = f(g(x))$ then $h'(x) = f'(g(x)) \cdot g'(x)$.

$f(x)$

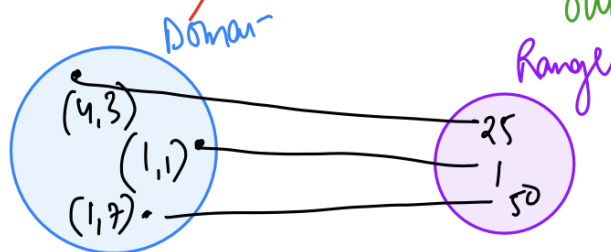
$\uparrow$ input variable.

$$f(x, y) = x^2 + y^2$$

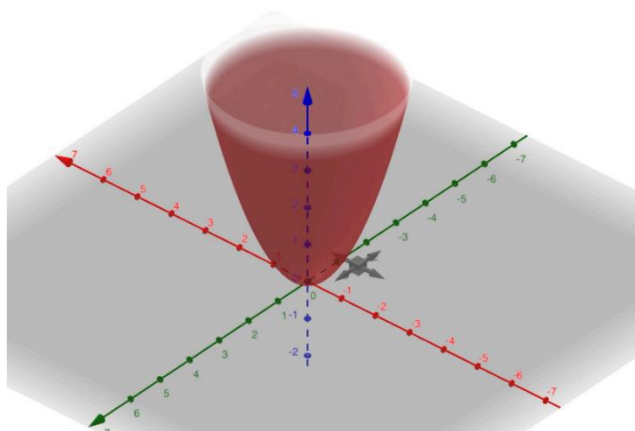
2 inputs

"multiple variables"

"Functions with multiple variables accept vectors as input & returns a single value as output."



$$f(x, y) = x^2 + y^2$$



Partial derivative

$$z = f(x, y) = x^2 + y^2$$

When we are taking Partial derivatives, we treat all other variables as constant.

$$z = f(x, y) = x^2 + y^2$$

$$\frac{\partial (z)}{\partial x} = 2x$$

$$\frac{\partial (z)}{\partial y} = 2y$$

$$Q) f(w_1, w_2, w_0) = w_1 x_1 + w_2 x_2 + w_0$$

$$(a) \frac{\partial f(w_1, w_2, w_0)}{\partial w_1} = x_1$$

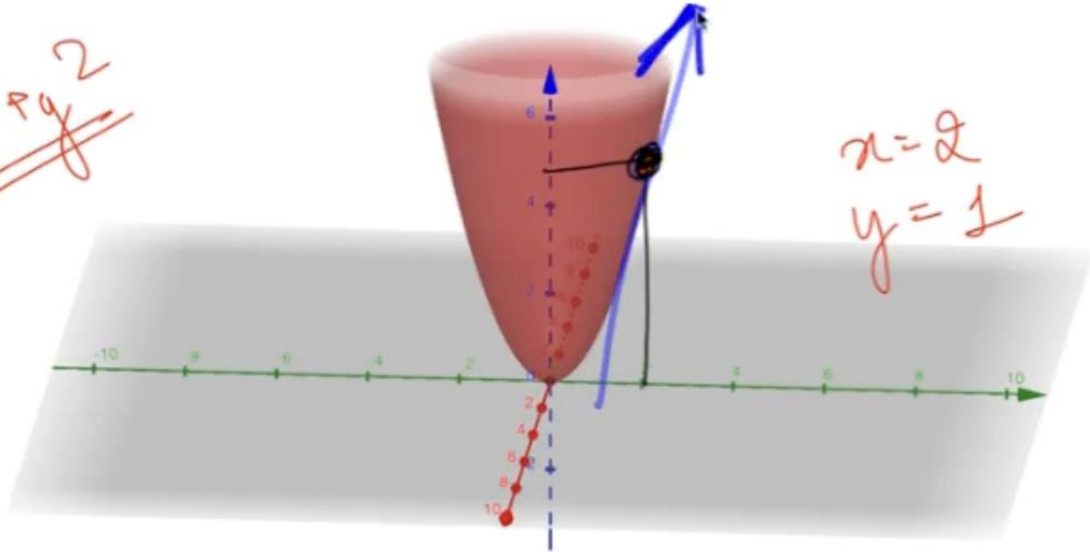
$$(b) \frac{\partial f(w_1, w_2, w_0)}{\partial w_2} = x_2$$

$$(c) \frac{\partial f(w_1, w_2, w_0)}{\partial w_0} = 1$$

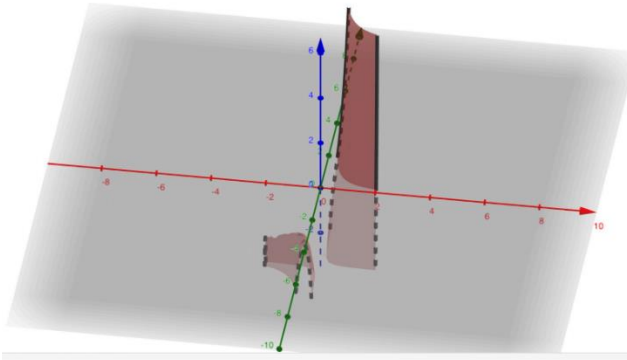
$$\nabla_{\bar{w}} f(\bar{w}) = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_0 \end{bmatrix} \quad f(w_1, w_2, w_0) = f(\bar{w})$$

~~$x^2 + y^2$~~



$x=2$
 $y=1$



$$f(x, y) = 3 \ln(xy) + 4y^2x^3$$

$$\nabla_{\bar{x}} f(\bar{x}) \quad \bar{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\nabla_{\bar{x}} f(\bar{x}) = \begin{bmatrix} \frac{\partial f(x, y)}{\partial x} \\ \frac{\partial f(x, y)}{\partial y} \end{bmatrix}$$

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &= 3 \cdot \frac{1}{xy} \cdot y + 4 \cdot y^2 \cdot 3x^2 \\ &= \frac{3}{x} + 12y^2x^2 \end{aligned}$$

$$\begin{aligned} \frac{\partial f(x, y)}{\partial y} &= 3 \cdot \frac{1}{xy} \cdot x + 4x^3 \cdot 2y \\ &= \frac{3}{y} + 8x^3y \end{aligned}$$

$$\nabla_{\bar{x}} f(\bar{x}) = \begin{bmatrix} \frac{3}{x} + 12y^2x^2 \\ \frac{3}{y} + 8x^3y \end{bmatrix}$$

Gradient of a vector.

derivative = single variable functions

Gradient = Multiple variable functions

$$y = x^2 = f(x)$$

$$f'(x) = 2x$$

$$\frac{d}{dx} f(x) = f'(x)$$

$$f(x, y)$$

$$f(\bar{x})$$

$$\bar{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\nabla_{\bar{x}} f(x)$$

Compute Gradient of any function

$$\nabla_{\bar{x}} f(\bar{x}) =$$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial}{\partial x_1} f(\bar{x}) \\ \frac{\partial}{\partial x_2} f(\bar{x}) \end{bmatrix}$$

$$Q) \quad f(x_1, x_2) = \log(x_1) + \frac{x_1}{x_2} e^{x_1+x_2}$$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\nabla_{\bar{x}} f(\bar{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} f(\bar{x}) \\ \frac{\partial}{\partial x_2} f(\bar{x}) \end{bmatrix}$$

$$\frac{1}{x_2} \cdot e^{x_1+x_2}$$

$$\frac{\partial}{\partial x_1} f(\bar{x}) = \frac{1}{x_1} + \frac{1}{x_2} \left[x_1 \cdot e^{x_1+x_2} \cdot 1 + e^{x_1+x_2} \right]$$

$$= \frac{1}{x_1} + \frac{e^{x_1+x_2}}{x_2} (x_1 + 1)$$

$$\frac{\partial}{\partial x_2} f(\bar{x}) = x_1 \left[-\frac{1}{x_2^2} e^{x_1+x_2} + \frac{1}{x_2} e^{x_1+x_2} \cdot 1 \right]$$

$$= \frac{x_1 (e^{x_1+x_2})}{x_2} \left[1 - \frac{1}{x_2} \right]$$

$$f(x) = x^2$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

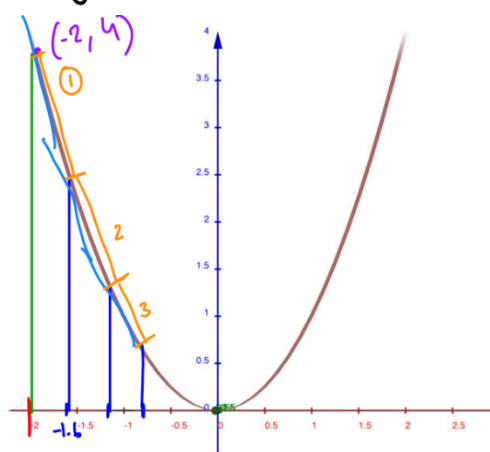
$$f(x, y) = x^2 + y^2$$

$$\frac{\partial}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{\partial}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Intuition
for
Gradient
Descent.

$y=4$



$$f(x) = x^2$$

$$f'(x) = 2x$$

$$f'(-2) = 2(-2) = -4$$

Step 1: Make an initial guess for x where our minima lies

Step 2: Find slope of tangent of function at that point.

Step 3: Update the guess using GD rule.

$$\begin{aligned} \eta &= 0.1 \\ \rightarrow x^{(1)} &= -2 - 0.1(-4) = -2 + 0.4 \\ &= \underline{-1.6} \\ \rightarrow x^{(2)} &= x^{(1)} - \eta \frac{d}{dx} f'(x) \\ x^{(2)} &= -1.6 - 0.1(-3.2) = -1.6 + 0.32 = -1.28 \end{aligned}$$

Our guess $x^{(0)}$. learning rate

$$x^{(1)} = x^{(0)} - \eta \frac{d}{dx} f(x)$$

Gradient Descent update Rule.

$$\mathcal{L}(X, \bar{w}, w_0) = - \sum_{i=1}^n \left(\frac{\bar{w}^T \bar{x}_i + w_0}{\|\bar{w}\|} \right) y_i$$

$$\nabla_{\bar{w}} \mathcal{L}(X, \bar{w}, w_0)$$

D-dimension.

$$\bar{w}^{(t+1)} = \bar{w}^{(t)} - \eta \nabla_{\bar{w}} \mathcal{L}(X, \bar{w}, w_0)$$

$$x^{(t+1)} = x^{(t)} - \eta \frac{d}{dx} f(x)$$

1 dimension.